SIMATS ENGINEERING

ASSESSMENT TOOL 3 TECHNICAL PRESENTATION

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5 **Total Marks:100**

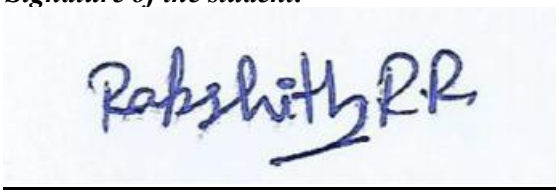
Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

*I **RAKSHITH R.R (192324156)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

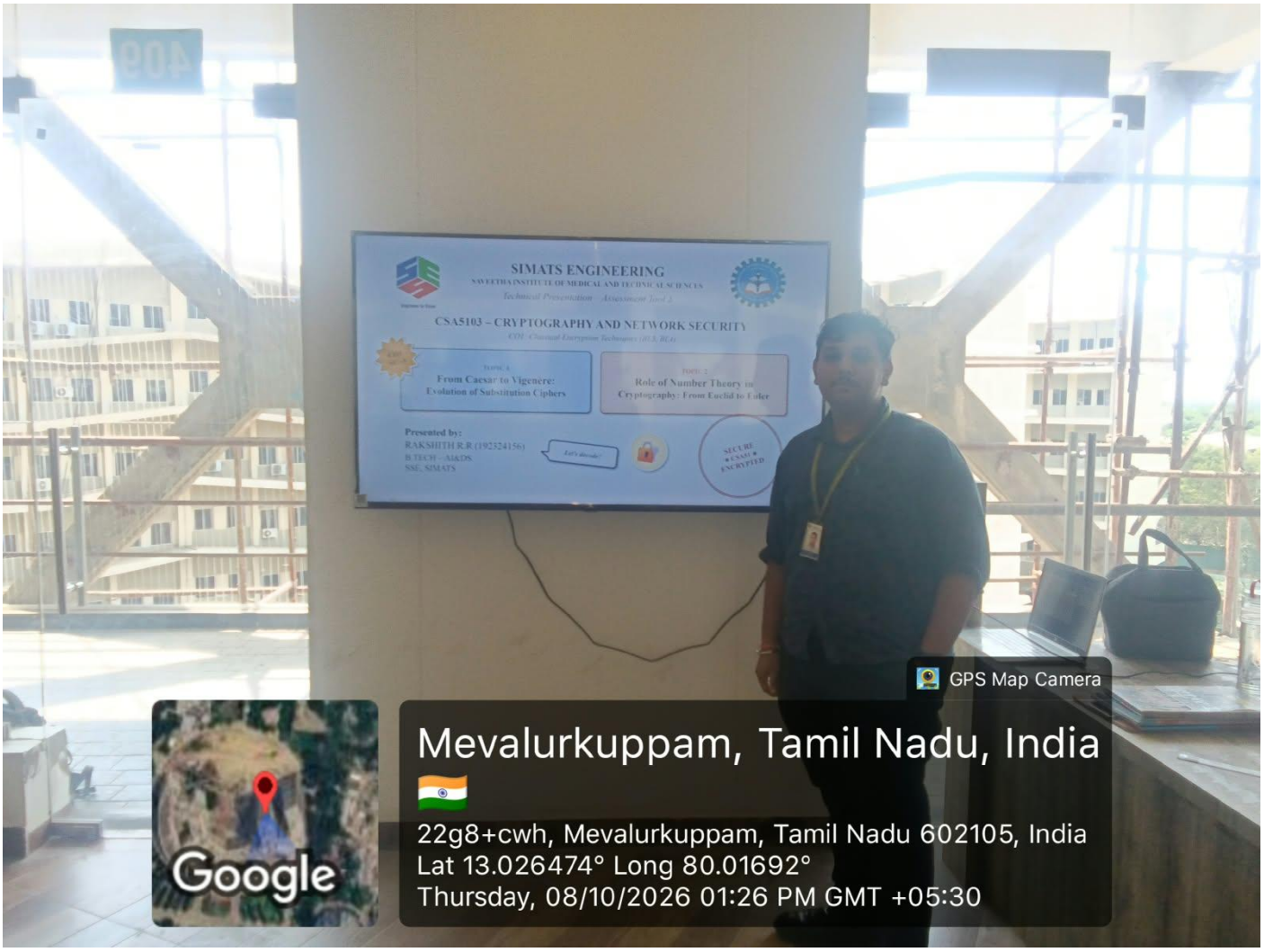
TECHNICAL PRESENTATION

1. From Caesar to Vigenère: Evolution of Substitution Ciphers”
2. “Substitution vs Transposition: Which Provides Better Security?”
3. “Security Attacks, Threats, Risks and Their Impact on Security Goals”
4. “Steganography vs Cryptography: Hidden Security in the Digital World”
5. “Role of Number Theory in Cryptography: From Euclid to Euler”

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding g; limited or partial integration	Poor understanding g; unable to integrate concepts
Experiment al Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation on with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

TECHNICAL PRESENTATION GEOTAG PHOTO



TECHNICAL PRESENTATION ALLOTED QUESTIONS

5.“Role of Number Theory in Cryptography: From Euclid to Euler”

1.“From Caesar to Vigenère: Evolution of Substitution Ciphers”

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

Technical Presentation – Assessment Tool 3

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CSA5103 – CRYPTOGRAPHY AND NETWORK SECURITY

CO1: Classical Encryption Techniques (BL3, BL4)

CO1
AT – 3

TOPIC 1

From Caesar to Vigenère:
Evolution of Substitution Ciphers

TOPIC 2

Role of Number Theory in Cryptography: From Euclid to Euler

Presented by:

RAKSHITH R.R (192324156)

B.TECH – AI&DS

SSE, SIMATS

Let's decode!



SECURE
★ CSA51 ★
ENCRYPTED

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AGENDA & OBJECTIVES

1

TOPIC 1 · Substitution Ciphers

- Caesar cipher and its weaknesses
- Evolution: monoalphabetic → polyalphabetic
- Vigenère cipher: working with an example
- Comparison and cryptanalysis

2

TOPIC 2 · Number Theory

- Euclid: gcd and modular inverse
- Fermat and Euler: φ(n) and the theorems
- RSA worked example with Python check
- Modern applications and security basis

Learning objectives (CO1)

- Apply classical ciphers to encrypt and decrypt messages.
- Analyse why classical ciphers fail and how mathematics fixes it.
- Verify every result by hand and by a short Python program.



CAESAR & THE BIRTH OF SUBSTITUTION CIPHERS

- **Substitution cipher:** each plaintext letter is replaced by another symbol; the order of letters does not change.
- **Caesar cipher:** every letter is shifted by a fixed key k (Julius Caesar used k = 3).
- **Encrypt:** `C = (P + k) mod 26`
- **Decrypt:** `P = (C − k) mod 26`
- **Weakness 1:** only 25 useful keys, so brute force takes seconds.
- **Weakness 2:** letter frequencies are preserved (E stays the commonest letter), so frequency analysis works.

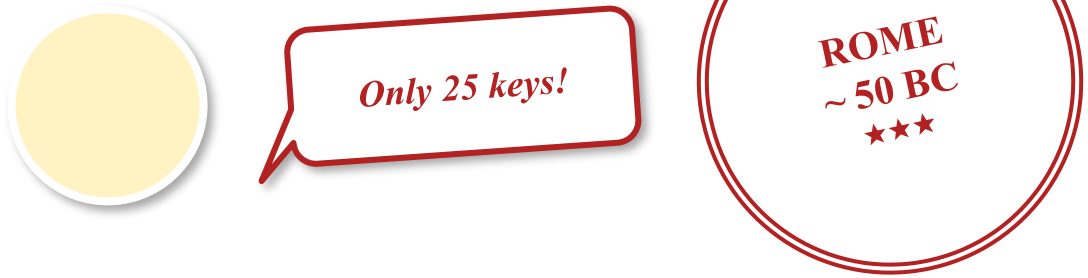
Plain	A	B	C	D	E	F	G	H	I	J	K	L	M
Cipher	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**

Plain	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**	**A**	**B**	**C**

Cipher alphabet for shift k = 3

HELLO → KHOOR

plaintext → ciphertext (k = 3)



EVOLUTION OF SUBSTITUTION CIPHERS

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MONO → POLY

~50 BC	9th century	1553	1854	1863	
1	2	3	4	5	6
Caesar cipher	**Al-Kindi**	**Bellaso / Vigenère**	**Playfair**	**Kasiski test**	**Vernam, rotor machines**
One fixed shift; only 25 keys	First frequency analysis breaks monoalphabetic ciphers	Keyword gives many shifts: polyalphabetic	Encrypts pairs of letters (digraphs)	Finds the Vigenère key length	Random key (one-time pad); Enigma

- Monoalphabetic
- Polyalphabetic / stronger
- Cryptanalysis (attack)

Every attack forced a stronger cipher

VIGENÈRE CIPHER IN ACTION

Step	1	2	3	4	5
Plaintext	**H**	**E**	**L**	**L**	**O**
P value (A=0)	7	4	11	11	14
Repeating key (KEY)	**K**	**E**	**Y**	**K**	**E**
K value	10	4	24	10	4
P + K	17	8	35	21	18
(P + K) mod 26	17	8	9	21	18
Ciphertext	**R**	**I**	**J**	**V**	**S**

Encrypt: $C = (P + K) \bmod 26$

Decrypt: $P = (C - K) \bmod 26$

K repeats: K, E, Y, K, E ...

Why it is stronger

The two L's become J and V: the same letter gets different cipher letters.

Python check ✓

```
def vig(p, k):
    return "".join(chr((ord(c) + ord(k[i % len(k)]) - 130) % 26 + 65)
                   for i, c in enumerate(p))
print(vig("HELLO", "KEY"))   # RIJVS
```

HELLO → RIJVS

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CAESAR vs VIGENÈRE: SECURITY ANALYSIS

Feature	Caesar	Vigenère
Key	One shift k (0–25)	Keyword of length m
Key space	25 keys	26^m (m = 5 → 11,881,376)
Letter frequencies	Preserved: peaks stay visible	Flattened across m alphabets
Typical attack	Brute force, frequency analysis	Kasiski / index of coincidence, then column-wise frequency analysis
Security today	**None**	**Broken (teaching use only)**

Lesson: a short repeating key always leaks its length.

Breaking Vigenère

1. Find key length m (Kasiski or index of coincidence)
2. Split the text into m columns
3. Run frequency analysis on each column

BROKEN
KASISKI
1863

EUCLID: GCD & MODULAR INVERSE

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- **Euclid's algorithm:** gcd(a, b) = gcd(b, a mod b); repeat until the remainder is 0.
- **Extended Euclid:** writes gcd = a·x + b·y, giving the inverse a^{-1} mod n when gcd = 1.
- **Affine cipher:** C = (aP + b) mod 26 needs gcd(a, 26) = 1, which leaves 12 values of a and 12 × 26 = 312 keys.
- **RSA:** the private key d is found with extended Euclid.

c. 300 BC
Euclid's
Elements

gcd = 1
✓



Example: find 7^{-1} mod 40

Forward (divide)	Backward (isolate remainder)
40 = 5 × 7 + 5	5 = 40 − 5×7
7 = 1 × 5 + 2	2 = 7 − 1×5
5 = 2 × 2 + 1	1 = 5 − 2×2

$1 = 3\times40 - 17\times7 \Rightarrow 7^{-1} \equiv -17 \equiv 23 \pmod{40}$

Check: 7 × 23 = 161 = 4×40 + 1 ✓

No factoring needed to find gcd!

FERMAT & EULER: THE KEY THEOREMS

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1640

FERMAT'S LITTLE THEOREM

If p is prime and p ∤ a:

a^(p−1) ≡ 1 (mod p)

Example: $3^6 = 729 = 104 \times 7 + 1$, so $3^6 \equiv 1 \pmod 7$

Use: fast primality tests and key generation

1763

EULER'S TOTIENT φ(n)

φ(n) = count of numbers 1..n coprime to n

$\varphi(p) = p - 1$

$\varphi(pq) = (p-1)(q-1)$

Example: $\varphi(15) = 2 \times 4 = 8$

1763

EULER'S THEOREM

If gcd(a, n) = 1:

a^φ(n) ≡ 1 (mod n)

Example: $2^8 = 256 = 17 \times 15 + 1$, so $2^8 \equiv 1 \pmod{15}$

Use: proves why RSA decryption works

Fermat's theorem is the special case of Euler's theorem with n = p, since φ(p) = p − 1.

$e \cdot d \equiv 1 \pmod{\varphi(n)}$

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RSA: NUMBER THEORY AT WORK



RSA 1977

Step	Computation
1	Pick primes p = 5, q = 11 ⇒ n = 55
2	φ(n) = (5−1)(11−1) = 40
3	Choose e = 7, since gcd(7, 40) = 1
4	$d = 7^{-1} \bmod 40 = 23$ (extended Euclid)
5	Public key (55, 7); private key d = 23
6	Encrypt m = 8: $8^7 \bmod 55 = 2$
7	**Decrypt: $2^{23} \bmod 55 = 8$ ✓**

Python verification

```
p, q = 5, 11
n, phi = p*q, (p-1)*(q-1)
e = 7
d = pow(e, -1, phi)    # 23
c = pow(8, e, n)       # 2
m = pow(c, d, n)       # 8
```

Security: an attacker sees n = 55 but must factor it to get φ(n). Real RSA uses n of 2048 bits.

ROUND TRIP
8 → 2 → 8



Euclid + Euler = RSA

NUMBER THEORY IN MODERN CRYPTOGRAPHY

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System	Number-theory tool	Used for	Security rests on
RSA	φ(n), modular inverse, modular exponentiation	Encryption, signatures	Factoring n = pq is hard
Diffie–Hellman	Primes, primitive roots, mod p powers	Key exchange (TLS)	Discrete logarithm problem
Elliptic curves (ECC)	Arithmetic on curves over mod p	Mobile, TLS, Bitcoin	Elliptic-curve discrete log
Miller–Rabin / Fermat test	Fermat's little theorem	Generating large primes	Fast probabilistic primality test
Caesar, Vigenère, affine	Only mod 26 arithmetic and gcd	Teaching, history	**Nothing hard: small key space**

From mod 26 to 2048-bit primes: same mathematics, much bigger numbers and genuinely hard problems.





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CONCLUSION & KEY TAKEAWAYS

Topic 1 · Substitution Ciphers

- Caesar uses one shift; Vigenère uses many, hiding letter frequencies.
- A short repeating key still leaks its length (Kasiski, index of coincidence).
- Lesson: security needs a huge key space and no patterns.

Topic 2 · Number Theory

KEY POINTS

- Euclid gives gcd and modular inverses.
- Fermat and Euler give φ(n) and the proof that RSA decryption works.
- Hard problems (factoring, discrete log) give modern security.

MISSION COMPLETE ★★★

References

- W. Stallings, Cryptography and Network Security, Pearson.
- D. Kahn, The Codebreakers, Scribner.
- A. Menezes, P. van Oorschot, S. Vanstone, Handbook of Applied Cryptography, CRC Press.

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THANK YOU



Questions & Discussion

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ENCRYPTED
✓ DELIVERED
✓

WKDQN BRX = THANK YOU (Caesar, k = 3)

Any Questions ?